Executable Evaluation Contracts for Target-Weight Trading Pipelines

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Abstract

A sequence of target weights is not a complete backtest specification. Reported performance also depends on when signals become executable, what residual cash earns, how turnover incurs cost, and whether the comparator bears the same exposure. We encode these choices as executable contracts in QUANTCORTEX, with a canonical target tape, fail-closed invariants, explicit cash and cost accounting, and content-hashed result artifacts.

We first use a retrospective six-ETF rotation case to show how an exact daily identity localizes a negative result: net CAGR is 1.40%, conventional cash-excess Sharpe is -0.15, and active risky allocation contributes -3.38% per year. We then run a protocol frozen in the public repository before the two input panels were retrieved. The expansion evaluates time-series momentum, cross-sectional momentum, short-term reversal, and a walk-forward gradient-boosted model over 2,011 sessions in U.S. sector and country-equity ETF panels. Across the resulting 8 family–panel cells, removing 13-basis-point costs raises annualized arithmetic return by 0.57–1.81 percentage points; all 8 pointwise 95% block-bootstrap intervals lie above zero. Setting residual-cash return to zero produces intervals below zero in 5 cells. Deliberately assigning close-derived targets to the return ending at that close produces intervals below zero in 3 cells and changes 4 within-panel family ranks. Engine conventions create a maximum daily return difference of 93.21 basis points.

These are controlled evaluation effects, not evidence of profitable alpha. The historical panels are neither temporally out of sample nor openly redistributable, and the intervals are pointwise rather than multiplicity-adjusted. The result is a reproducible evaluation methodology and reference implementation with an explicit boundary around its empirical claims.

1 Introduction

A target-weight vector is a convenient interface between prediction, portfolio construction, and execution. Qlib and FinRL use modular research pipelines [Yang et al., 2020, Liu et al., 2020]; FinRL-X extends the interface through broker execution [Yang et al., 2026]. A target, however, is an intention rather than a realized return. A complete evaluation must also state when that target can be held, what uninvested capital earns, how turnover becomes cost, which holdings are credited with each return interval, and what happens when an order outcome is uncertain.

These choices correspond to separate hypotheses. The information set may be causal while the cash account is incomplete. The accounting may be correct while the code is nondeterministic. A reproducible backtest may still offer no investment value, and a valid simulation says little about whether retrying an uncertain broker request is safe. Treating these questions as one notion of "correctness" obscures the source of failure.

The distinction is particularly important in financial machine learning, where observations are dependent, effective samples are limited, and repeated strategy search creates substantial false-discovery

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risk [White, 2000, Harvey et al., 2016, Bailey and López de Prado, 2014, Bailey et al., 2017, Arnott et al., 2019]. We ask a narrower systems question: how can a target-weight pipeline make its economic and operational semantics executable, testable, and traceable from the information set to paper-order state?

The primary contribution is the evaluation methodology; the strategies are test vehicles and the software is a reference implementation. Specifically, we (i) define a machine-readable evaluation contract and canonical target tape for information timing, weights, execution lag, cash, costs, comparison, and uncertain order outcomes; (ii) formalize an exact daily return decomposition for changing risky exposure and preserve it under joint block resampling; (iii) estimate five one-switch effects under a repository-frozen protocol across four heterogeneous strategy families and two real-data panels; and (iv) bind every reported result to its protocol, clean source revision, environment, input digest, and artifact hash. The purpose is to expose sensitivity to evaluation semantics, not to select the best-performing strategy after inspection.

2 Related work

Financial backtest validity. Sharpe-ratio inference depends on distributional and serial-dependence assumptions [Lo, 2002]. Data snooping, repeated trials, and researcher degrees of freedom can make a selected backtest appear stronger than the process that generated it [White, 2000, Bailey and López de Prado, 2014, Bailey et al., 2017, Harvey et al., 2016]. Arnott et al. [2019] organize these risks into a research protocol. In a recent preprint, Zhang et al. [2026] estimate decision-time leakage by changing one evaluation convention at a time across multiple models and panels. Our same-close diagnostic is narrower: it is one contract test inside a broader accounting and execution audit, not a general leakage benchmark.

Attribution and matched comparison. Performance evaluation has long separated investment decisions into interpretable return components [Fama, 1972, Brinson and Fachler, 1985]. We apply that principle to a portfolio whose risky exposure changes over time. The identity separates passive exposure, exposure timing around the sample mean, active allocation, and modeled cost. It is an exact arithmetic decomposition, not a new geometric linking method. The strategy set draws on cross-sectional and time-series momentum, short-term reversal, residual momentum, and gradient boosting [Jegadeesh and Titman, 1993, Moskowitz et al., 2012, Lehmann, 1990, Blitz et al., 2011, Friedman, 2001], but each model is used only as a fixed test vehicle.

Research software and implementation risk. Qlib and FinRL provide modular financial-AI workflows [Yang et al., 2020, Liu et al., 2020]; FinRL-X emphasizes a shared weight interface across research and deployment [Yang et al., 2026]. Computational reproducibility remains difficult even with code and data. Across 1,000 computational reproduction attempts in finance, the original code regenerated the reported result exactly only 52% of the time [Pérignon et al., 2024]. In a recent preprint, Yin et al. [2026] compare five engines across 15 strategies and find material divergence from implementation choices, including transaction costs. Our two-engine diagnostic is deliberately more limited: it compares two documented holding conventions on identical target tapes, not independent third-party implementations or an estimate of industry-wide implementation risk. The manifest and source fingerprints follow the broader reproducibility objectives of Pineau et al. [2021]. Our contribution lies in connecting those artifacts to explicit economic contracts within one executable evaluation. No individual invariant or accounting identity is claimed as new; the contribution is their executable integration and controlled stress test.

3 Executable audit protocol

We use "contract" in the design-by-contract sense: an executable invariant with an explicit failure response [Meyer, 1992]. Figure 1 summarizes the evaluation path. Each stage can fail independently, and a later stage does not repair an invalid earlier one. Table 1 states the checks used in this study.

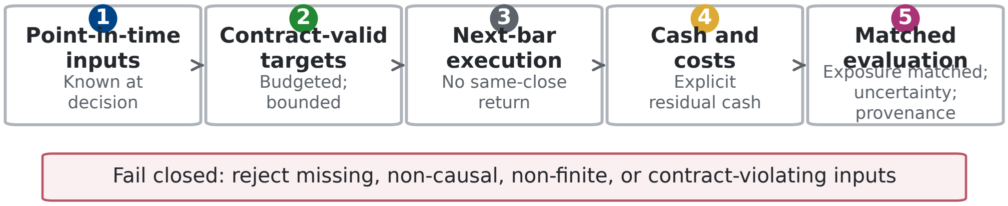

Figure 1: Audit flow from point-in-time inputs to matched evaluation. The audited path validates required inputs, timing, finiteness, and contracts at the stage where each condition becomes testable.

Table 1: Core invariants and failure behavior.

Invariant	Operational check	Failure behavior
Information	Features and fundamentals must be observable by the decision timestamp.	Reject future-dated merges; do not backfill.
Allocation	Allocator output is finite, bounded, and budget-consistent.	Raise a contract error.
Overlay	Timing and risk overlays may reduce exposure but cannot create an invalid book.	Reject non-finite or over-limit weights.
Timing	A close-derived target earns returns only after the first strictly later execution bar.	Shift holdings; never use the observed close twice.
Accounting	Risky assets, residual cash, turnover, and costs share one capital clock.	Reject missing cash bars or cost models.
Comparison	Exact attribution uses realized exposure; a tradable control follows target exposure and pays costs.	Label ex-post controls; charge costs to implementable comparators.
Execution	Persist intent before broker submission; uncertain outcomes are not retried automatically.	Block until broker reconciliation.

3.1 Portfolio, execution, and cash contracts

Let $w_d \in \mathbb{R}^N$ be the risky-asset target decided after observing the close at decision time d, and let $e(d)$ be the first available execution bar strictly after d. At $e(d)$, the primary event engine solves adjusted-close pseudo-shares $q_{i,e(d)}$ against post-cost NAV so that the resulting risky allocation matches w_d. It holds q_i fixed between rebalances. If $P_{i,t}$ is the adjusted close and V_t is portfolio NAV, the realized risky weight at the prior close is $h_{i,t-1} = q_{i,t-1}P_{i,t-1}/V_{t-1}$. Close-derived targets are therefore not filled at the close they observe and do not earn the return ending at their execution close. For the long-only case, $h_{i,t} \geq 0$ and $\sum_i h_{i,t} \leq 1$; the residual cash holding is

$$h_t^c = 1 - \sum_{i=1}^{N} h_{i,t}. \tag{1}$$

For simple asset returns $r_{i,t}$ over $(t-1, t]$ and cash-account return r_t^c, pre-cost portfolio return is therefore

$$r_t^{\text{gross}} = \sum_{i=1}^{N} h_{i,t-1} r_{i,t} + h_{t-1}^c r_t^c. \tag{2}$$

Missing cash observations fail the run instead of silently becoming zero. For executed risky-notional change $\Delta a_{i,t}$ measured against pre-trade NAV, define $B_t = \sum_i \max(\Delta a_{i,t}, 0)$ and $S_t = \sum_i \max(-\Delta a_{i,t}, 0)$. One-way turnover is $\max(B_t, S_t)$, gross traded notional is $B_t + S_t$, and the charge as a fraction of pre-trade NAV is $\kappa_t = 0.0013(B_t + S_t)$. If V_t^- is pre-trade NAV and V_{t-1} is prior-close NAV, the return drag is $d_t = \kappa_t V_t^- / V_{t-1}$ and $r_t^{\text{net}} = r_t^{\text{gross}} - d_t$. Target shares are solved against post-cost NAV. The cost model does not separately estimate commissions, bid–ask spread, taxes, or market impact. This proportional model is a first-order sensitivity assumption;

large or urgent orders require models that account for order size, volatility, and execution urgency, such as Almgren and Chriss [2001].

We report nominal CAGR, maximum drawdown, annualized one-way turnover, annualized gross traded notional, and the conventional sample cash-excess Sharpe statistic

$$\mathrm{SR}_c = \sqrt{252}\,\frac{\mathrm{mean}(r_t - r_t^c)}{\mathrm{sd}(r_t - r_t^c)}. \tag{3}$$

Equation 3 is a descriptive statistic, not an iid-normal inference procedure; serial dependence is represented approximately in a separate circular-block sensitivity analysis [Lo, 2002]. An SHV adjusted-close series supplies the return credited to residual cash; the engine does not trade SHV when cash weight changes. It is a short-duration cash proxy, not a frictionless risk-free asset. Because the input contains adjusted closes, the pseudo-shares are total-return accounting units with implicit distribution and corporate-action adjustment, not broker-faithful nominal shares.

3.2 Exact return attribution

Headline performance does not identify which decision created or destroyed value. Let $x_t = \sum_i h_{i,t-1}$ be risky exposure during return period t, let b_t be the return of an equal-initial-weight buy-and-hold basket, and let $\bar{x}$ be the full-sample mean of x_t. Define the daily exposure-matched passive return and the constant-exposure diagnostic as

$$m_t = x_t b_t + (1 - x_t) r_t^c, \tag{4}$$

$$k_t = \bar{x} b_t + (1 - \bar{x}) r_t^c. \tag{5}$$

Then net excess over cash has the exact arithmetic decomposition

$$\begin{aligned} r_t^{\mathrm{net}} - r_t^c = {} & \underbrace{(r_t^{\mathrm{gross}} - m_t)}_{\text{active risky allocation}} + \underbrace{(m_t - k_t)}_{\text{dynamic exposure timing}} \\ & + \underbrace{(k_t - r_t^c)}_{\text{passive risky exposure}} + \underbrace{(r_t^{\mathrm{net}} - r_t^{\mathrm{gross}})}_{\text{implementation cost}}. \end{aligned} \tag{6}$$

The first term includes group selection, within-risky-asset weighting, and any return effect of the engine's between-rebalance weight semantics; it is not pure security selection in the Brinson sense. The timing term measures daily exposure changes relative to the sample-average exposure. Because $\bar{x}$ is estimated over the full evaluation window, k_t is an ex-post diagnostic rather than a tradable ex-ante benchmark. Equation 6 holds on every date, so arithmetic annualized means add exactly. CAGR does not admit the same additive interpretation.

We resample all five series in Equation 6 with common circular-block indices. Consequently, the identity also holds within every bootstrap draw. The reported fraction of positive draws is a resampling diagnostic, not a posterior probability or a multiple-testing-adjusted p-value.

3.3 Single-assumption diagnostics

Three diagnostics change one assumption while holding the targets, primary engine, and input matrix fixed: residual cash earns zero, proportional costs are removed, or a close-derived target is deliberately executed one bar before that close so it earns the return ending at the observed close. The last construction uses future information and is therefore invalid. Under explicit share holdings, changing the cash return can also change realized exposure between rebalances through the NAV denominator. These are sensitivity checks, not candidate strategies, and they play no role in model selection. The paired design resembles the one-switch logic of Zhang et al. [2026], but here the switches audit cash, costs, and execution semantics in one pipeline.

3.4 Reference implementation and order-state controls

The reference implementation separates data, alpha, portfolio, timing, risk, backtest, execution, and strategy layers. Allocator outputs satisfy a strict budget contract. Post-overlay outputs use a relaxed exposure contract because a timing or risk gate may intentionally leave capital in cash. Point-in-time merge guards reject fundamentals that were not yet public, and optional broker, machine-learning, and provider SDKs remain lazy imports so the scientific core can be tested offline.

A central manifest records the clean source revision, Python and package versions, dependency lock, input SHA-256, experiment design, source-tree hashes, and the hash of every published table and figure. A versioned JSON contract and canonical long-form target-tape schema make the engine boundary machine readable. Each empirical run round-trips every strategy decision stream through the versioned payload boundary and records a canonical payload hash; deterministic fixtures test the same boundary without using the empirical price matrix. The empirical sections below exercise the research and backtest contracts. Order-state controls are verified by deterministic tests: paper-trading state is written atomically as a versioned snapshot, intent is persisted before submission, and an uncertain submission is not retried automatically. SDK-conformance tests instantiate the pinned Alpaca and Interactive Brokers request types. They do not test authenticated transport or venue behavior. Appendix G separates exact identities, property tests, counterfactuals, fault injection, SDK conformance, and untested live behavior.

4 Evaluation design

4.1 Retrospective worked audit

The first study is a fixed six-ETF weekly rotation strategy evaluated from 2018-01-02 through 2025-12-31. It combines group ranking, residual momentum, a Gaussian-mixture exposure gate, and volatility scaling. The study was assembled before a prospective protocol existed, and its earlier trial count is unknown; it is therefore a worked accounting audit, not confirmatory evidence. Its known construct issues are retained rather than repaired after seeing the result: QQQ is both a group member and the ranking benchmark, two regime features are exact volatility rescalings, early regime windows are immature, and one seed controls the mixture fit. Appendix A gives the complete configuration.

The owner-supplied adjusted-close matrix is not distributed. It is identified by a manifest-recorded SHA-256. The event engine credits SHV to residual cash, charges 13 basis points per dollar traded on each buy or sell leg, and executes close-derived weekly targets on the first strictly later bar. The exact attribution uses a realized-exposure ex-post control; a separate causal target-exposure comparator pays the same modeled costs.

4.2 Repository-frozen expansion

The second study was specified in the public repository before its two provider requests. The freeze is a Git timestamp, not an external registry entry, and the historical interval is not temporally out of sample. Two complete-row panels use adjusted closes from 2014-01-02 through 2025-12-31: nine U.S. sector ETFs and ten country-equity ETFs, each with SHV. Evaluation covers 2,011 sessions from 2018-01-02 through 2025-12-31. There is no forward fill or symbol substitution. The panels contain surviving, currently available ETFs and share broad market exposures, so they are not independent replications.

Table 2 summarizes the frozen strategy families. Each decision uses the first complete session of the month and executes on the first strictly later panel close. Long-only target exposure is at most one; unused capital earns SHV. The primary event engine holds adjusted-close pseudo-shares between rebalances and charges 13 basis points per dollar traded. Every target round-trips through the canonical JSON tape before evaluation.

The learned family pools asset-month observations without symbol identity. Features are 5-, 21-, 63-, 126-, and 252-session log returns; 21- and 63-session sample volatility; distance from the trailing 252-session high; and ordinal cross-sectional ranks of 21- and 252-session returns. They are neither standardized nor winsorized. At each evaluation decision the model uses only training labels whose 21-session endpoint is observable by that close, retains at most 60 mature decision months, and requires at least 24. A gradient-boosted regressor with 100 depth-2 trees, learning rate 0.03, minimum leaf size 10, and subsample 0.8 is refit separately for seeds 11, 29, 47, 71, and 97. Features and labels are recomputed walk-forward; no fitted model crosses a decision boundary.

Table 2: Frozen expansion strategies. Ties use ascending symbols.

Family	Fixed rule
Time-series momentum	Positive 252-session total returns; equal weight across eligible assets.
Cross-sectional momentum	Top three returns from $t - 252$ to $t - 21$; one-third each.
Short-term reversal	Up to three lowest negative 5-session returns; one-third each, without renormalizing unused exposure.
Learned GBRT	Walk-forward 21-session return prediction from return, volatility, trailing-high, and rank features; top three positive predictions; five frozen seeds.

4.3 One-switch estimands and uncertainty

For each family–panel cell, five paired switches compare (i) invalid same-close assignment with next-bar execution, (ii) zero cash return with SHV, (iii) zero cost with 13-basis-point cost, (iv) the strategy with a causal costed equal-weight comparator at the same target risky exposure, and (v) vectorized with event-driven accounting. The comparator is conditional on the strategy's target exposure and is a conditional exposure control, not an independent market benchmark. The vectorized engine re-pegs weights between decisions, whereas the event engine holds drifting pseudo-shares; their difference is a model-convention sensitivity, not automatically an implementation error.

Primary outcomes are paired changes in annualized arithmetic return and conventional SHV-excess Sharpe. We use 5,000 joint circular-block draws with a 21-session primary block and repeat the calculation with 5 and 63 sessions [Politis and Romano, 1992]. Return series, cash, and all five learned seeds share each draw. Intervals are unstudentized pointwise 95% percentile intervals. They condition on the selected panels and frozen models, are not multiplicity adjusted, and do not establish future performance.

5 Results

5.1 Baseline outcomes are inputs to the audit, not discoveries

Appendix Figure 5 reports after-cost arithmetic return and SHV-excess Sharpe. 7 of 8 family–panel cells have positive sample Sharpe; country-panel time-series momentum is slightly negative. The sector panel is stronger for every family. These rankings are descriptive: the panels, families, and 2018–2025 interval are not a random sample, and no baseline interval or multiple-testing correction supports an alpha claim.

5.2 Contract effects depend on the strategy

Table 3 counts pointwise intervals across the eight family–panel cells. Cost removal is a positive-by-construction check, but its size is material: annualized arithmetic return rises by 0.57–1.81 percentage points. In contrast, same-close assignment is not a uniformly optimistic bias. Its interval is below zero in 3 cells and overlaps zero in 5, driven by the deliberately contrarian reversal rule and, in one panel, the learned model. Zero cash return is below zero in 5 cells and overlaps zero in 3. These patterns show why a convention cannot be summarized by one global correction factor.

Table 3: Number of 21-session pointwise intervals below zero / overlapping zero / above zero, out of eight family–panel cells. Counts are descriptive and are not multiplicity-adjusted tests.

Switch difference	Annual return	Sharpe
Same-close - next-bar	3/5/0	3/5/0
Zero cash - SHV cash	5/3/0	5/3/0
Zero cost - 13 bp cost	0/0/8	0/0/8
Strategy - costed comparator	1/7/0	1/7/0
Vectorized - event-driven	0/4/4	0/4/4

Figure 2 displays the return effects. Relative to its costed exposure-matched comparator, only country-panel time-series momentum has an interval below zero; the other seven overlap zero. This

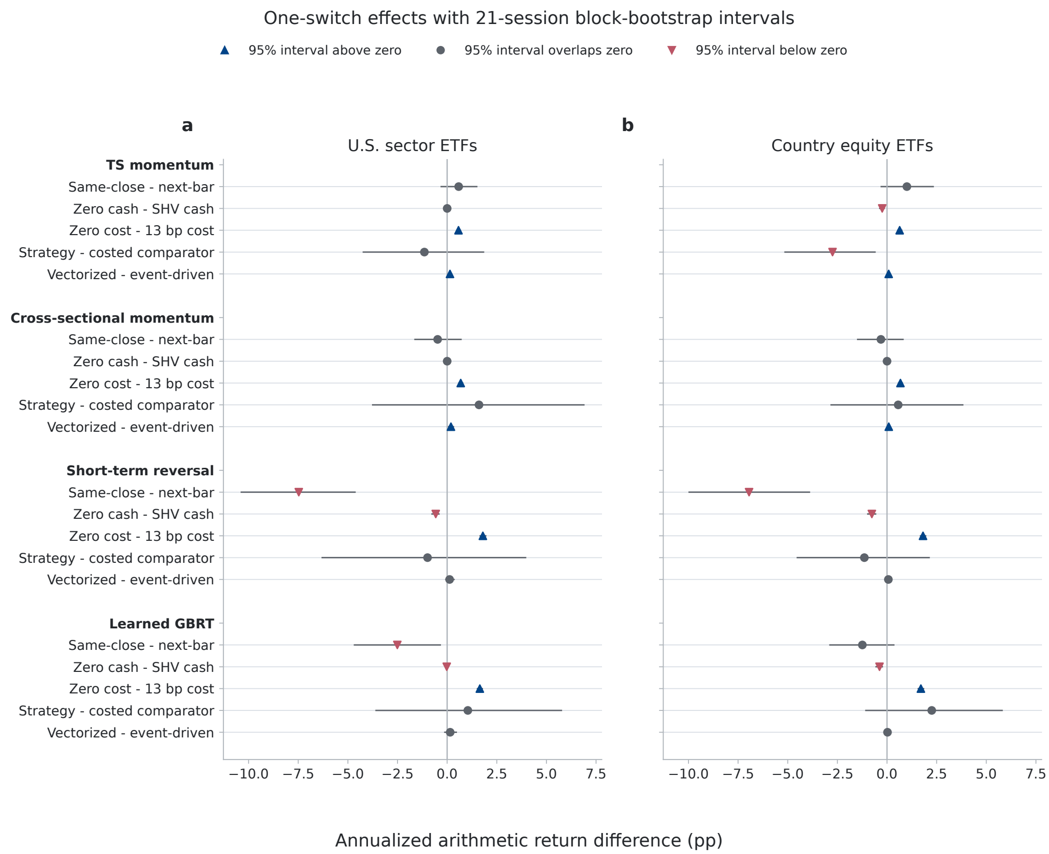

Figure 2: Paired annualized arithmetic-return effects with 21-session circular-block percentile intervals. Marker shape and color encode whether the pointwise interval is above, overlaps, or is below zero. The same-close switch is causally invalid and is never a candidate strategy.

does not validate the strategies. It says the conditional comparator is not precise enough to rank most cells on this single path. Same-close and zero-cost switches change 4 and 4 of the 8 within-panel family ranks, respectively; replacing strategies with their costed comparators changes 5 ranks.

5.3 Seed and engine sensitivity remain visible

The learned family is not represented by a favorable seed. Across the five frozen seeds, baseline SHV-excess Sharpe ranges from 0.47–0.60 in sectors and 0.27–0.37 in countries; every seed-level result is published. The vectorized-minus-event mean effect is small, but the maximum one-day difference reaches 93.21 basis points and the largest absolute terminal wealth gap is 3.89%. This divergence is expected from daily weight re-pegging versus drifting pseudo-shares. Appendix Figures 7 and 8 report both diagnostics.

5.4 The worked audit localizes one negative result

The retrospective case records 1.40% net CAGR and -0.15 SHV-excess Sharpe. Its causal target-exposure comparator, after its own costs, records 5.72% and 0.62; annualized arithmetic shortfall is -4.14% [-6.70%, -1.63%]. Equation 6 attributes net excess over SHV exactly: passive risky exposure contributes 3.90%, active risky allocation -3.38%, dynamic exposure timing 0.30%, and modeled cost -1.72% per year. The active-allocation interval [-5.85%, -0.90%] is wholly below zero.

Crediting SHV instead of zero to residual cash raises CAGR from -0.17% to 1.40% because mean risky exposure is only 30.14%. Omitting modeled costs changes Sharpe from -0.15 to 0.14; invalid same-close assignment changes it to 0.03. The expansion qualifies that last result: look-ahead changes estimates and ranks, but its direction depends on the signal. Appendix A.1 contains accounting, cost, subperiod, and block-length diagnostics for the worked case.

6 Limitations and broader impact

6.1 Statistical validity

The retrospective strategy predates the protocol, and no complete log of its earlier trials survives. The expansion protocol was committed before the two provider requests, but it was not externally registered and evaluates an historical interval. Neither study is a temporal holdout. A Deflated Sharpe Ratio or Reality Check would require a defensible count of prior trials that is not available [White, 2000, Bailey and López de Prado, 2014].

The eight family–panel cells are dependent: models share dates, panels share global market exposures, and five learned seeds are repeated fits rather than independent strategies. The circular-block bootstrap conditions on these paths and approximates serial dependence through fixed block lengths. Pointwise intervals and interval-category counts are not multiplicity adjusted. Stability across 5, 21, and 63 sessions is sensitivity evidence, not proof of stationarity or future performance. The conventional $\sqrt{252}$ Sharpe remains a descriptive statistic; block resampling does not remove the assumptions needed to interpret it. Component interval endpoints need not add, although the exact decomposition holds within every joint draw.

6.2 Economic construct validity

"Active risky allocation" in the worked case combines group selection, within-risky weighting, and between-rebalance engine semantics. "Dynamic exposure timing" is relative to full-sample mean exposure, hence diagnostic rather than tradable. In both studies, the exposure-matched comparator is conditional on the strategy's own target path; it controls exposure but is not an independent benchmark. Arithmetic means decompose exactly; compounded wealth does not.

The frozen models are deliberately simple. The ETF panels do not represent point-in-time constituent universes, the country funds are unhedged U.S.-listed securities, and the learned model pools symbols without identity or hyperparameter search. SHV has duration, expenses, and market-price variation; it is not a risk-free account. Adjusted-close pseudo-shares embed total-return adjustments and omit broker share records, spreads, queue position, and fill uncertainty. The proportional cost model has no order-size, volatility, venue, impact, ADV, or capacity term.

6.3 External validity and reproducibility

The expansion broadens the worked case but still covers two U.S.-listed ETF panels, four hand-specified families, one eight-year interval, and two internal engines. It does not estimate a population distribution of backtest errors or match broader multi-model and multi-engine studies [Zhang et al., 2026, Yin et al., 2026]. Provider-adjusted histories can be revised. Raw inputs are not redistributed, and provider permission for the derived publication has not been independently verified. Open schemas, canonical tapes, and conformance fixtures reproduce software semantics, not the reported returns; full empirical reproduction requires owner-supplied matrices with the recorded hashes.

6.4 Operational scope and broader impact

Fault-injection tests cover stale or uncertain execution state, but no live authentication, venue behavior, funded order, capacity, or production control was tested. The intended benefit is reduced false confidence; the principal risk is mistaking an audited simulation for investment advice or deployment approval. Paper-mode defaults and fail-closed retries mitigate but do not remove that risk. Past simulated performance does not imply future returns.

7 Conclusion

A target-weight generator does not determine a realized return; information timing, execution, cash, costs, and comparator exposure complete the experiment. In the retrospective case, exact attribution shows that active allocation and modeled cost erase the benefit of passive risky exposure. In the frozen expansion, removing modeled cost raises annualized return in every family–panel cell, while cash, same-close timing, comparator, and engine effects vary by strategy. The same-close switch can worsen a contrarian model rather than mechanically inflate all results; the error is using unavailable information, not a guaranteed sign of bias. Canonical tapes, one-switch counterfactuals, and artifact provenance make those distinctions testable. The contribution is an executable evaluation methodology and bounded negative evidence, not a profitable signal or deployment claim.

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A Retrospective case specification

The universe is QQQ, VGT, GLD, TLT, SPY, and VIG, grouped as growth equities, gold and nominal duration, and broad and dividend equities. On the first session of each week, the strategy keeps the top two groups by a 126-session information ratio relative to QQQ. Within selected groups, it estimates a one-factor regression over 126 sessions and ranks a separate 126-session residual-return window with a 21-session gap. Positive scores receive proportional long-only weights capped at 60% per asset.

A seeded three-component Gaussian mixture maps QQQ return and two exact rescalings of realized volatility to zero, half, or full exposure. A separate 21-session volatility scaler targets 20 volatility points. Named variants remove the regime gate, the volatility scaler, or both; none was selected after the result. The full manifest records every parameter and warm-up requirement.

A.1 Retrospective accounting results

Table 4: Audited retrospective accounting over 2018–2025.

Series	CAGR	Sharpe	Max drawdown
Strategy, after costs and SHV cash	1.40%	-0.15	-9.83%
Strategy, before costs and SHV cash	3.17%	0.14	-8.60%
SHV cash proxy	2.50%	–	-0.45%
Realized-exposure control, gross	6.74%	0.80	-4.64%
Target-exposure comparator, after costs	5.72%	0.62	-4.94%
Strategy, after costs and zero cash	-0.17%	0.00	-15.15%

Table 5: Exact attribution of annualized arithmetic net excess over SHV in the retrospective case. Intervals use common 21-session block draws.

Component	Mean	95% interval	Positive draws
Active risky allocation	-3.38%	[-5.85%, -0.90%]	15/5000
Dynamic exposure timing	0.30%	[-2.57%, 3.26%]	2889/5000
Passive risky exposure	3.90%	[0.82%, 6.85%]	4963/5000
Modeled implementation cost	-1.72%	[-2.08%, -1.38%]	0/5000
Net excess over cash	-0.90%	[-4.77%, 3.14%]	1682/5000

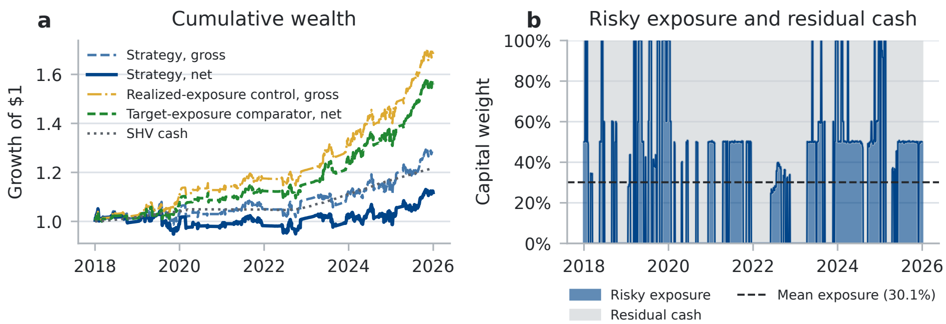

Figure 3: Growth of one dollar and the capital split between risky assets and SHV under the audited retrospective accounting.

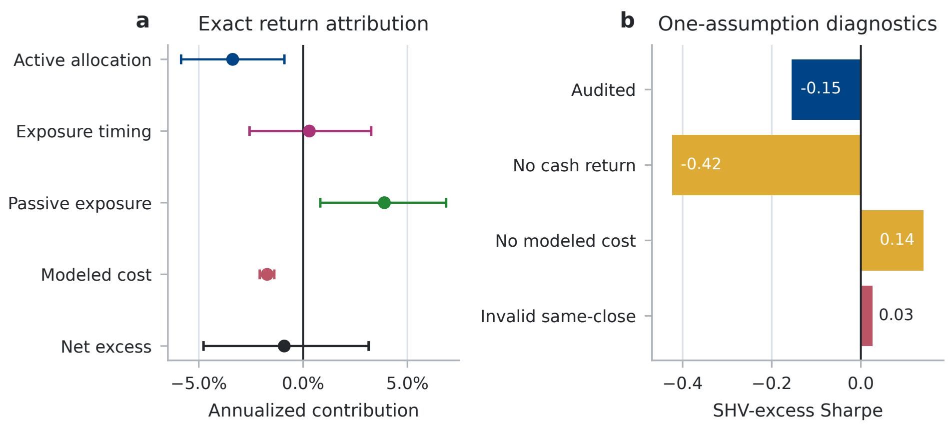

Figure 4: Exact return attribution and one-assumption diagnostics for the retrospective case. The same-close result is causally invalid.

B Reproducibility details

The retrospective and expansion generators are `scripts/run_paper_experiments.py` and `scripts/run_expansion_experiments.py`. Their release wrappers check out the recorded source revision in a detached clean worktree, keep raw inputs outside that worktree, and verify source cleanliness and output hashes. The reviewed invocations are:

```
scripts/release_expansion_artifacts.sh \
  local_data/expansion
scripts/release_paper_artifacts.sh \
  local_data/published_rotation_prices.csv
```

Exact empirical reproduction requires three owner-supplied files matching the recorded SHA-256 values: the six-ETF retrospective matrix and the two expansion panels. Aggregate CSVs, generated LaTeX, figures, and hashes are listed in `paper/results/manifest.json` and `paper/expansion/results/manifest.json`. The expansion manifest also records the pre-data protocol commit, provider-request metadata, canonical tape hashes, thread libraries, dependency lock, and every source file used by the run. The three matrix digests are:

Retrospective: `efb384a62157e56a0cd8065abf45c1ed07d90ec26c681e5d54d74fe4cb9c55e1`
U.S. sectors: `c5947c4ca6ad6d21ad834c8f344dcdd07acc59ba411e3dcb2202a2413642b2f9`
Country equities: `870fa926b5080378c65bd629b9959bd73b16391dded5720c894c0f35558132a9`

The expansion protocol digest is

`e49e41a12a19fa5404a573ba5e21eb8a2888e616985f8c610d9652866923315c`

The frozen JSON key `cost_per_one_way_gross_notional` is misnamed: its 0.0013 value is applied to gross two-sided traded notional, as defined in Section 3. The key is retained to preserve the pre-retrieval protocol hash. The retrospective source-tree digest is:

`7dfddd78bc3d88c01b39666be0157aa9e5cb728c9def9bd8f911ce03f4716e27`

The anonymous review build omits author and repository identifiers. Both reviewed runs were CPU-only; environment versions and thread-pool metadata are machine readable. Supported CI runs use Python 3.11 through 3.14. Repository-wide testing, packaging, notebook execution, and broker conformance checks are engineering validation rather than model search.

C Expansion tables and diagnostics

Panel	Family	Ann. return	Sharpe	CAGR	Max DD
U.S. sector ETFs	TS momentum	11.44%	0.49	10.29%	-34.16%
U.S. sector ETFs	Cross-sectional momentum	14.19%	0.60	13.04%	-32.06%
U.S. sector ETFs	Short-term reversal	11.02%	0.44	9.54%	-45.08%
U.S. sector ETFs	Learned GBRT	13.66%	0.53	12.10%	-39.96%
Country equity ETFs	TS momentum	2.18%	-0.02	0.88%	-40.45%
Country equity ETFs	Cross-sectional momentum	8.84%	0.33	7.21%	-40.73%
Country equity ETFs	Short-term reversal	6.03%	0.20	4.48%	-43.91%
Country equity ETFs	Learned GBRT	7.84%	0.31	6.56%	-33.56%

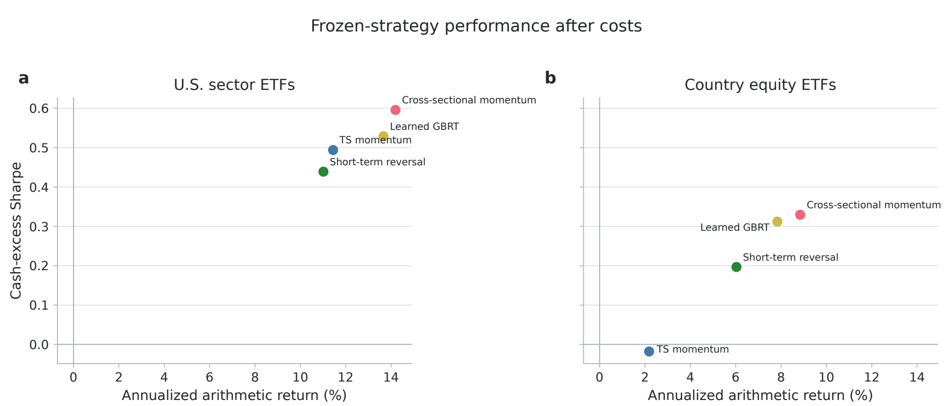

Figure 5: After-cost baseline estimates under the frozen expansion protocol. The learned point is the arithmetic mean across five seed-level metrics.

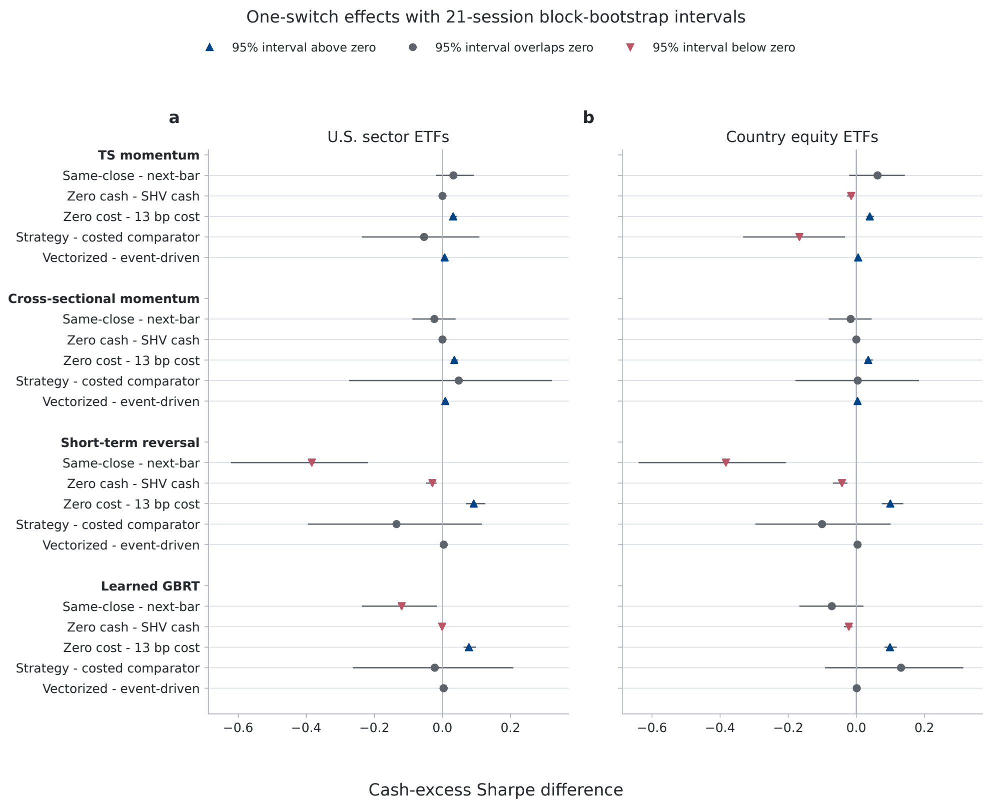

Figure 6: Paired conventional cash-excess Sharpe effects and 21-session pointwise block-bootstrap intervals for all expansion cells.

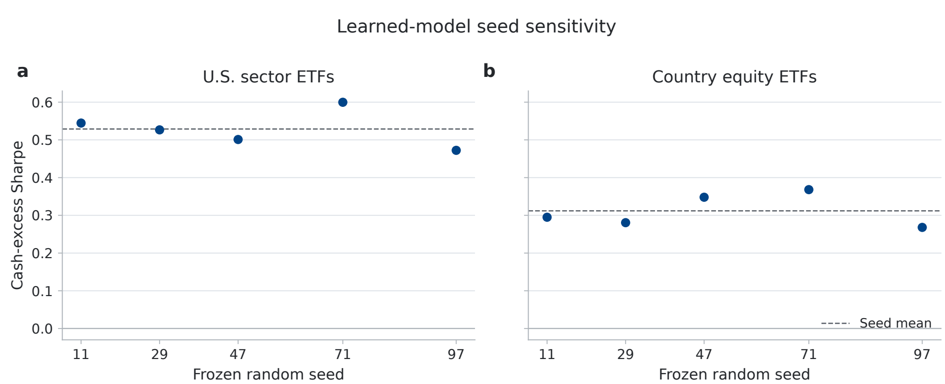

Figure 7: Learned-model baseline Sharpe for all five frozen random seeds. Dashed lines are within-panel seed means.

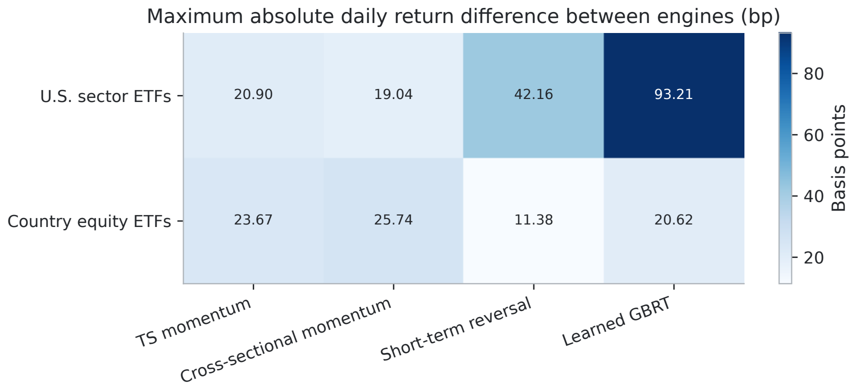

Figure 8: Maximum absolute one-day return difference between event-driven pseudo-share accounting and the vectorized daily re-pegging convention.

D Retrospective block-length sensitivity

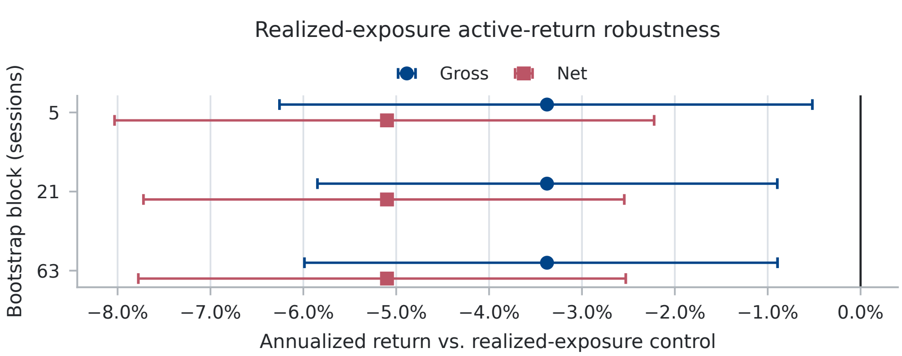

Figure 9: Active-return estimates and 95% percentile intervals under 5-, 21-, and 63-session circular-block resampling against the realized-exposure attribution control. Gross compares before-cost strategy returns with the gross control; net retains modeled strategy costs while the control remains gross.

E Retrospective engine comparison

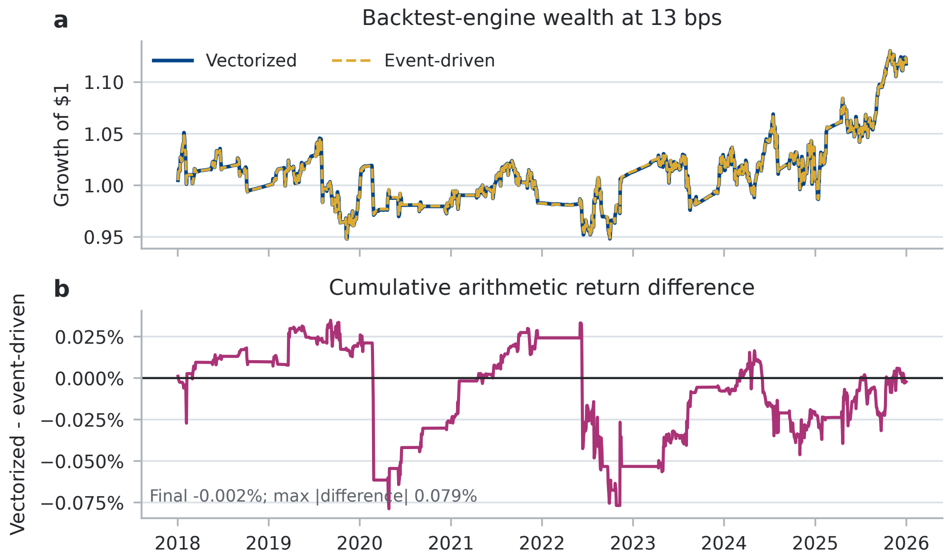

Figure 10: The vectorized engine re-pegs target weights between rebalances, whereas the event-driven engine holds explicit drifting adjusted-close pseudo-shares. Their small difference here does not remove the semantic distinction.

F Retrospective cost and overlay sensitivity

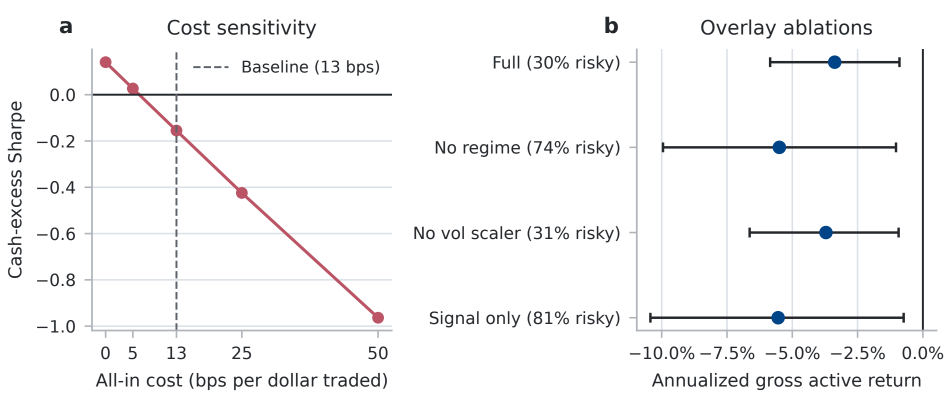

Figure 11: (a) After-cost Sharpe sensitivity for the full strategy. (b) Annualized gross strategy return minus the realized-exposure attribution control, with 95% intervals from common 21-session circular-block draws. Tick labels report mean risky exposure; all named audit variants are shown.

G Contract-to-evidence map

The artifact distinguishes evidence types rather than treating every passing check as equivalent:

Evidence class	Claim supported	Committed evidence
Exact identity	Daily return components reconstruct net cash excess	Per-row assertion and `return_decomposition.csv`
Property test	Weight, timing, cash, and target-tape invariants hold	Deterministic pytest cases, including hand-computable tapes
Counterfactual	One declared assumption changes the reported metric	`protocol_switches.csv` and `contract_effects.csv`
Fault injection	Unsafe or stale execution state fails closed	Truncated reconciliation, unknown status, and stale-writer tests
SDK conformance	Offline request and response mappings match pinned SDKs	Broker model-construction tests and behavioral mocks
Untested live	Authenticated transport, venue state, and funded orders	No supporting evidence; explicitly outside scope

H Use of language-model assistance

The author used LLM-based coding assistants for implementation planning, repository inspection, code drafting, adversarial review, regression tests, and copyediting. The assistants are not an evaluated model or data source. The author approved and publicly froze the expansion protocol before its provider requests, reviewed the code and artifacts, and remains responsible for the methods and claims. Numerical claims are generated by committed scripts and machine-readable artifacts.

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